

Project handbook  
VoteVox  
1.5

Version 1.5  
Project manager: Hubauer Paul Matthias  
Date: 08.12.2025

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# Document versions

| Versionno. | Date       | Change                               | Author                                                        |
|------------|------------|--------------------------------------|---------------------------------------------------------------|
| 1          | 13.11.2025 | File created & first data filled out | Hubauer Paul Matthias<br>Aigner Sandro                        |
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|            |            |                                      |                                                               |
|            |            |                                      |                                                               |



# 1 Project plans

## 1.1 Project Assignment

| <div>VoteVox</div> <div>1.1</div> <div>PROJECT-ASSIGNMENT</div>                                                                                                                                                                                                              |                                                                                                                                                                                                                           |           |                       |      |           |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------|------|-----------|--|--|--|
| <b>Project start event:</b> <ul style="list-style-type: none"> <li>First team meeting</li> </ul>                                                                                                                                                                             | <b>Project start date:</b> <ul style="list-style-type: none"> <li>12<sup>th</sup> March 2026</li> </ul>                                                                                                                   |           |                       |      |           |  |  |  |
| <b>project close down event in terms of content:</b> <ul style="list-style-type: none"> <li>Completion of the program and successful tests</li> </ul> <b>Formal project close down event:</b> <ul style="list-style-type: none"> <li>Release of the first version</li> </ul> | <b>Project close down dates:</b> <ul style="list-style-type: none"> <li>Content Completion: 7th May 2026</li> <li>Formal Closing: 13th May 2026</li> </ul>                                                                |           |                       |      |           |  |  |  |
| <b>Project objectives:</b> <ul style="list-style-type: none"> <li>QR-based voting</li> <li>Admin Panel to create the codes</li> <li>Detailed analytics of the results</li> </ul>                                                                                             | <b>Non-objectives:</b> <ul style="list-style-type: none"> <li>Multi language support (only english planned)</li> <li>Mobile application (web-only for the voters)</li> <li>History data comparison</li> </ul>             |           |                       |      |           |  |  |  |
| <b>Main tasks (Project phases):</b> <ul style="list-style-type: none"> <li>Secure, anonymous &amp; fair voting</li> <li>Result statistics and analytics for admins</li> <li>Easy to use website for voters</li> </ul>                                                        | <b>Project resources and costs*:</b> <table border="1"> <thead> <tr> <th>resource/type of cost</th> <th>unit</th> <th>Costs (€)</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table> |           | resource/type of cost | unit | Costs (€) |  |  |  |
| resource/type of cost                                                                                                                                                                                                                                                        | unit                                                                                                                                                                                                                      | Costs (€) |                       |      |           |  |  |  |
|                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                           |           |                       |      |           |  |  |  |
| <b>Project owner:</b> <ul style="list-style-type: none"> <li>Hubauer Paul Matthias</li> </ul>                                                                                                                                                                                | <b>Project manager:</b> <ul style="list-style-type: none"> <li>Hubauer Paul Matthias</li> </ul>                                                                                                                           |           |                       |      |           |  |  |  |
| <b>Project team members:</b> <ul style="list-style-type: none"> <li>Anzengruber Laurenz</li> <li>Aigner Sandro</li> </ul>                                                                                                                                                    |                                                                                                                                                                                                                           |           |                       |      |           |  |  |  |
| <div> <div>.....</div> <div>Hubauer Paul Matthias, (Project owner)</div> </div> <div> <div>.....</div> <div>Hubauer Paul Matthias (Project manager)</div> </div>                                                                                                             |                                                                                                                                                                                                                           |           |                       |      |           |  |  |  |

\* Possible categories of total Project budget:

Category A: up to 0,3 million Euro

Category B: up to 1 million Euro

Category C: up to 10 million Euro

Category D: more than 10 million Euro

## 1.2 Project Objectives (objectives, non-objectives)

| <div> <div>VoteVox</div> <div>1.2</div> <div>PROJECT OBJECTIVES</div> </div>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Type of objective                                                                                                                 | Project objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Adjusted project objectives as of...               |
| <b>objectives:</b> <ul style="list-style-type: none"> <li><b>Main objectives</b></li> <li><b>Additional objectives</b></li> </ul> | <ul style="list-style-type: none"> <li>Develop a secure, QR code-based digital voting system for school and department representative elections</li> <li>Implement one-time use authentication tokens to ensure each student can vote only once</li> <li>Create an intuitive voting interface accessible via web browsers on various devices</li> <li>Build a comprehensive admin panel for election management and real-time monitoring</li> <li>Design and implement a detailed results evaluation dashboard with analytics by class and department</li> <li>Generate exportable reports (PDF, Excel) for election documentation</li> <li>Create user documentation and admin guides for future elections</li> </ul> | <ul style="list-style-type: none"> <li></li> </ul> |
| <b>Non-objectives</b>                                                                                                             | <ul style="list-style-type: none"> <li>Integration with existing school management systems (might be added in a future version)</li> <li>Multi-language interface support (English only)</li> <li>Development of native mobile applications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li></li> </ul> |

## 1.3 Description of Pre- and Post Project Phase

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### DESCRIPTION OF PRE- AND POST- PROJECT PHASE

#### 1) Pre-project phase

*What triggered the project?*

- Need for a more efficient and transparent voting process for school and department representative elections
- Desire to modernize democratic processes and increase student engagement through digital solutions

*Relevant documents for the project („Minutes“, ... ONLY documents and no content necessary)*

- VoteVox\_Requirement\_Specification.pdf

*Experience from similar projects*

- Knowledge from web development projects in school context
- Experience with database design and user authentication from prior coursework

#### 2) Post-project phase

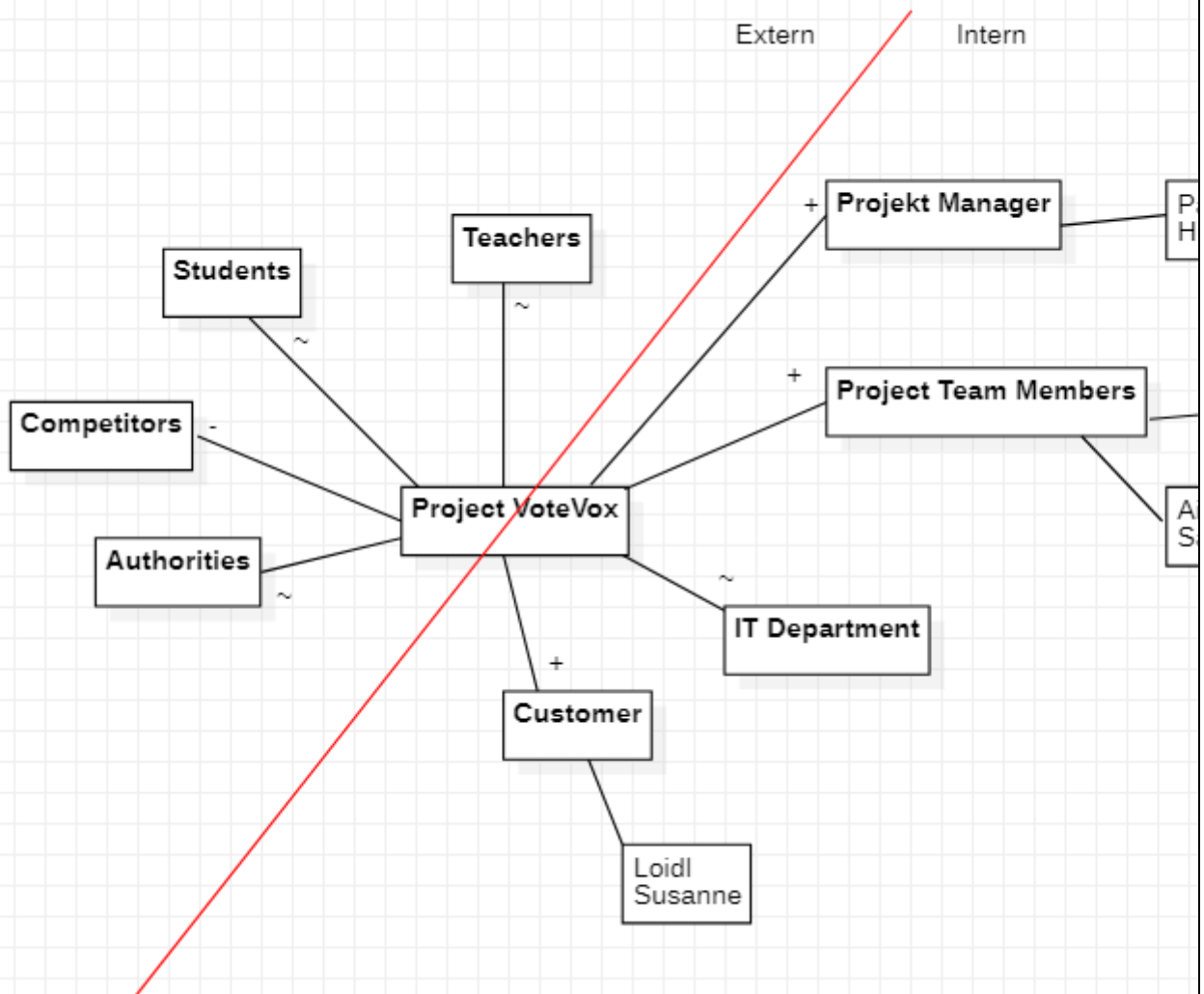
*What will happen after the project has ended? (follow-up activities, further projects, ...)?*

- Training session for school administration and election committee on system usage
- Collection of user feedback and system performance metrics during first live elections
- Documentation of lessons learned and areas for improvement
- Planning for VoteVox 2.0 with additional features (multi-language support, integration with school management system)
- Potential expansion to other schools in the region who express interest in the system

## 1.4 Project Environment Analysis

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### PROJECT ENVIRONMENT GRAPHIC





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## PROJECT ENVIRONMENT TABLE

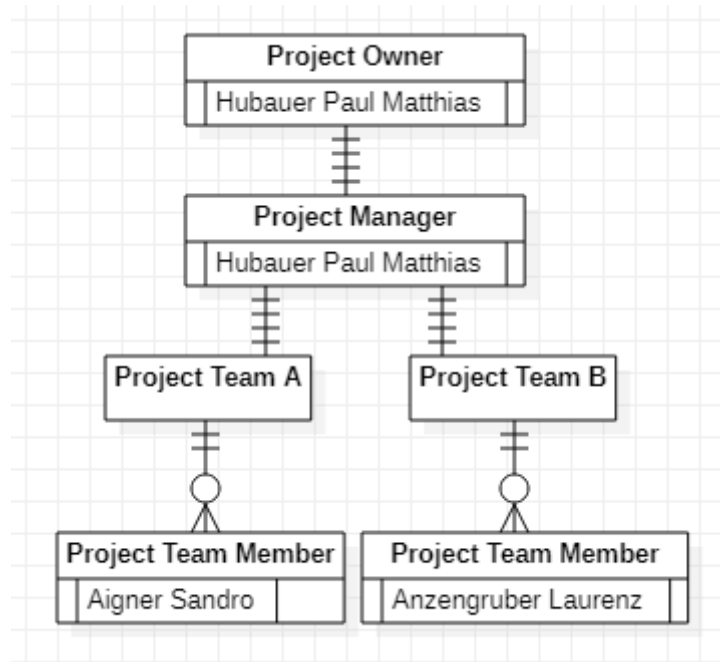
| Environment          | Relationship<br>(potential/conflict)                                                                                                                                                                                                    | Measures                                                                                                                                              | Who / when<br><br>PSP Code                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Project Manager      | <b>Potential:</b> Effective coordination, clear vision, timely decision-making<br><b>Conflict:</b> Overload from dual role (owner & manager), communication bottlenecks                                                                 | Weekly team meetings, clear task delegation, time management tools, escalation protocols, shared project dashboard                                    | Project Manager<br><br>Weekly                           |
| Project Team Members | <b>Potential:</b> Strong technical skills, collaborative work<br><b>Conflict:</b> Workload imbalance, miscommunication, merge conflicts, skill gaps                                                                                     | Daily standups, Git workflow (branch/merge strategy), pair programming sessions, code reviews, clear role boundaries                                  | Project Team Members<br><br>Weekly                      |
| Competitors          | <b>Potential:</b> Learn from existing solutions, understand market standards, feature inspiration<br><b>Conflict:</b> Pressure to match features, unrealistic comparisons, potential discouragement from established solutions          | Competitive analysis (Week 1-2), feature gap analysis, USP (Unique Selling Point) definition, focus on school-specific needs, avoid feature creep     | Project Team Members<br><br>Week 1 & 2                  |
| Authorities          | <b>Potential:</b> Ensures legal compliance, builds trust with stakeholders, protect data<br><b>Conflict:</b> Strict regulations may limit functionality, extensive documentation requirements, approval delays                          | Privacy impact assessment, data flow documentation, consent mechanism implementation, GDPR compliance checklist, legal review sessions                | Project Manager<br><br>Week 1 & 2                       |
| Customer             | <b>Potential:</b> Budget approval, resource allocation, future project opportunities<br><b>Conflict:</b> Budget constraints, changing requirements, unrealistic timelines                                                               | Bi-weekly status meetings, transparent budget tracking, formal change request process, milestone-based approvals, written communication for decisions | Project Manager<br><br>Bi-Weekly                        |
| IT Department        | <b>Potential:</b> Server infrastructure, technical expertise, security protocols, ongoing maintenance support<br><b>Conflict:</b> Strict security requirements, limited server resources, slow approval processes, competing priorities | Initial technical requirements meeting (Week 1), security audit (Week 5), infrastructure capacity planning, firewall/network access coordination      | Project Manager, Project Team Members<br><br>Week 1 & 5 |
| Teachers             | <b>Potential:</b> Support adoption, provide practical insights, classroom coordination<br><b>Conflict:</b> Limited time for training, skepticism about digital voting, concerns about technical issues during election                  | Teacher information session (Week 6), quick start guides, designated teacher champions, technical support hotline, backup procedures documentation    | Anzengruber<br>Laurenz<br><br>Week 6                    |
| Student              | <b>Potential:</b> High tech adoption, enthusiasm for modernization, valuable feedback, increased engagement<br><b>Conflict:</b> Privacy concerns, resistance to a new system, usability complaints                                      | Pilot testing with volunteer class (Week 6), anonymous feedback surveys, clear privacy explanations, user-friendly interface design, tutorial videos  | Aigner Sandro<br><br>Week 6                             |

## 1.5 Relationship to Other Projects and the Organisations' Strategy

| <div> VoteVox<br/>1.2 </div> <div>RELATIONSHIP TO OTHER PROJECTS</div> |                                      |           |                        |
|------------------------------------------------------------------------|--------------------------------------|-----------|------------------------|
| Programs/<br>Projects/                                                 | Relationship<br>(potential/conflict) | Sanctions | Who / when<br>WBS Code |
|                                                                        |                                      |           |                        |
|                                                                        |                                      |           |                        |
|                                                                        |                                      |           |                        |
|                                                                        |                                      |           |                        |
|                                                                        |                                      |           |                        |
|                                                                        |                                      |           |                        |

| <div> VoteVox<br/>1.2 </div> <div>CONNECTION TO THE ORGANISATION'S STRATEGY</div> |                                        |
|-----------------------------------------------------------------------------------|----------------------------------------|
| Strategy                                                                          | Description of connection/relationship |
|                                                                                   |                                        |
|                                                                                   |                                        |
|                                                                                   |                                        |
|                                                                                   |                                        |
|                                                                                   |                                        |
|                                                                                   |                                        |

## 1.6 Project Organisation Chart



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### PROJECT- ORGANISATION

| Role in Project      | Field of duties/Skills             | Name                               |
|----------------------|------------------------------------|------------------------------------|
| Project owner        | Set Goals and manage employees     | Hubauer Paul Matthias              |
| Project manager      | Set tasks and controll all workers | Hubauer Paul Matthias              |
| Project team members | Progress the tasks                 | Aigner Sandro, Anzengruber Laurenz |
| Project members      | -                                  | -                                  |

## 1.7 Plan of Objects of Consideration of the Project

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### PLAN OF OBJECTS OF CONSIDERATION

#### 1. Objectives

##### Primary Goal

- Development of a secure, intuitive, and digital voting system for school and department representative elections, ensuring a simple and one-time vote submission using QR-code authentication

##### Secondary Goals

- Maximize voter turnout through easy accessibility (web-based)
- Provide a Admin Panel for managing elections, candidates, and QR generation
- Automated creation of detailed statistical evaluations and export options (PDF, Excel/CSV) for election documentation
- Guarantee security and data integrity throughout the entire election process

#### 2. Target Groups

- **Eligible Students:** All students entitled to vote for the school or department representative
- **Election Commission/Administration:** Teachers and administrative staff responsible for the preparation (QR code generation), execution (managing the voting timeframe), and evaluation (statistics) of the elections
- **Candidates:** Students running for the office of school or department representative

#### 3. Functionalities

- **Preparation & QR Code Generation**
  - **Data Collection:** Recording of all eligible students (Department, Class)
  - **QR Code Creation:** Generation of a single-use and encrypted QR code per student, containing token, assignment, and eligibility.
  - **Output:** Export of the generated QR codes as PDF (individually and by class).
- **Voting Process**
  - **Authentication:** Students scan their personal QR code to validate voting eligibility and perform their vote
  - **Vote Type Display:** Automatic display of the permitted voting masks (School Representative Election, Department Representative Election) based on eligibility
  - **Vote Submission:** Intuitive voting interface for casting the vote
  - **Deactivation:** After vote submission, the used QR code/token is immediately marked as invalid
- **Technical Architecture & Data Privacy**
  - **Admin Panel:** Central interface for QR code generation, candidate management and election start/stop
  - **Database:** Storage of votes and metadata with anonymization
- **Evaluation Options**
  - **Detailed Statistics**
  - **Visualization:** Presentation of results through diagrams
  - **Export:** Functions to export results as a **PDF Report** and as **Excel/CSV** for further analysis.

#### 4. Technical Requirements

- **Platform**
  - **Web Voting UI:** The voting interface must be accessible via web browsers
  - **Desktop Management UI:** The admin interface is desktop based
- **Technology**
  - **Frontend:** Ensuring an intuitive and responsive voting and admin interface
  - **Backend & Database:** Use of a reliable and secure architecture for storing votes and metadata (e.g., PostgreSQL-based) with a focus on data integrity
  - **Security:** Implementation of encryption and hashing mechanisms for QR codes and votes

#### 5. Design & Quality

- **User Interface**
  - **Priority:** Simplicity, speed, and clarity to minimize errors during vote submission
  - **Intuitive Navigation:** Voting should only require a maximum of three to four clicks
- **Responsive Design**
  - **Device Independence:** Full support and flawless display on various screen sizes and devices
- **Tests & Quality**
  - **Usability Tests:** Conducting tests in a beta phase to verify user-friendliness
  - **Safety Tests:** Verification of security features (QR code uniqueness, tamper-proofing)
  - **Performance Tests:** Ensuring the system remains stable and fast even with high simultaneous access (peak times)

#### 6. Security Features

- **Single Use:** Each QR code/token is immediately invalid after the first vote submission
- **Time Window:** Limiting the voting period to a defined time window
- **Logging:** Maintaining a complete log of all relevant system actions
- **Backup Mechanism:** Implementing a reliable backup system to prevent data loss
- **Manipulation Detection:** Logic for detecting and logging suspicious activities or attempts at multiple voting

#### 7. Support & Documentation

- **Customer Support**
  - **Open Channels:** Provision of support channels for the election commission (e.g., email) during the election process
- **Documentation**
  - **User Guides:** Creation of clear, illustrated documentation for the election commission regarding QR code generation, candidate management, and election evaluation

## 1.8 Work Breakdown Structure (WBS)

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### WORK BREAKDOWN STRUCTURE (WBS)



VoteVox\_Work\_Break  
down\_Structure.pdf

## 1.9 1.Project Work-Package Specification

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                    |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>VoteVox<br/>1.4</p>                               | <h3>PROJECT WORK-PACKAGE SPECIFICATION</h3>                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                    |
| <p><b>0.1</b><br/><b>Coordinate Project</b></p>      | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>• Distribute the Work-Packages</li> <li>• Manage Time Errors</li> <li>• Keep the documentation up to date</li> </ul>                                                                                                                                                                                                                                         | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>• Smooth Completion of the project</li> </ul>                                                                                                       |
|                                                      | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>• Project Status report</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                    |
| <p><b>0.2</b><br/><b>Control Project</b></p>         | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>• Observe the flow of the project and make decisions accordingly</li> </ul>                                                                                                                                                                                                                                                                                  | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>• A completed final version of the project</li> </ul>                                                                                               |
|                                                      | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>• Project Status Report</li> </ul>                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                    |
| <p><b>1.1</b><br/><b>Requirements Analysis</b></p>   | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>• Finalize and document all functional and non-functional requirements based on the Plan of Objects of Consideration</li> <li>• Define the overall technical architecture, including technology stack (languages, frameworks, database)</li> <li>• Identify and document all legal and GDPR compliance requirements for data handling and storage</li> </ul> | <p><b>Non-WP Content</b></p> <ul style="list-style-type: none"> <li>• No design activities (WP 2.0)</li> <li>• No coding or environment setup</li> <li>• No procurement of external hosting or services</li> </ul> |
|                                                      | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>• A signed-off <b>Requirements Specification Document (RSD)</b></li> <li>• The <b>Technical Architecture Blueprint</b></li> </ul>                                                                                                                                                                                                                             | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>• Completion of the RSD draft</li> <li>• Final review and approval of the complete Project Plan and Architecture Blueprint</li> </ul>    |
| <p><b>2.1</b><br/><b>Voting Interface Design</b></p> | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>• Create low-fidelity wireframes for the entire student voting flow</li> <li>• Develop high-fidelity screens for all necessary device sizes</li> </ul>                                                                                                                                                                                                       |                                                                                                                                                                                                                    |
|                                                      | <p><b>Non-WP Content</b></p>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                    |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <ul style="list-style-type: none"> <li>No actual front-end coding or development</li> <li>No design work for the Admin Panel</li> </ul>                                                                                                                                                                                                                                                                                                           |
|                                       | <b>WP Result</b> <ul style="list-style-type: none"> <li>A complete design system for the student application (style guide)</li> </ul>                                                                                                                                                                                                                                                                                                             |
|                                       | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Completion of low-fidelity wireframes</li> <li>Completion and internal review of all high-fidelity screens</li> </ul>                                                                                                                                                                                                                                                          |
| <b>2.2<br/>Admin Interface Design</b> | <b>WP Content</b> <ul style="list-style-type: none"> <li>Create low-fidelity wireframes for the Admin Panel functionalities</li> <li>Develop high-fidelity mockups for all Admin Panel screens, maintaining consistency with the overall design system</li> <li>Design the user experience for complex tasks such as bulk QR code generation and report exports</li> </ul>                                                                        |
|                                       | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No actual front-end coding or development</li> <li>No design work for the Voting Interface</li> </ul>                                                                                                                                                                                                                                                                                |
|                                       | <b>WP Result</b> <ul style="list-style-type: none"> <li>A complete design system for the administrator application (style guide)</li> </ul>                                                                                                                                                                                                                                                                                                       |
|                                       | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Completion of low-fidelity wireframes</li> <li>Completion and internal review of all high-fidelity screens</li> </ul>                                                                                                                                                                                                                                                          |
| <b>3.1<br/>Database Setup</b>         | <b>WP Content</b> <ul style="list-style-type: none"> <li>Install and configure the selected database system</li> <li>Design and implement the database schema</li> <li>Configure security, access roles, and initial backup mechanisms</li> </ul>                                                                                                                                                                                                 |
|                                       | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No implementation of the core voting business logic</li> <li>No integration with the front-end applications</li> </ul>                                                                                                                                                                                                                                                               |
|                                       | <b>WP Result</b> <ul style="list-style-type: none"> <li>A fully operational and secured database instance</li> <li>The finalized <b>Database Schema Document</b></li> <li>Initial API or repository layer for data persistence</li> </ul>                                                                                                                                                                                                         |
|                                       | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Installation and security configuration</li> <li>Final schema design and implementation, including security checks</li> </ul>                                                                                                                                                                                                                                                  |
| <b>3.2<br/>Logic Implementation</b>   | <b>WP Content</b> <ul style="list-style-type: none"> <li>Implement the backend API endpoint for <b>QR code/token validation</b> and eligibility checking</li> <li>Develop and test the <b>one-time voting mechanism</b></li> <li>Implement the logic to correctly record votes for the two election types based on the token's embedded eligibility</li> <li>Ensure all vote records are <b>anonymous and immutable</b> once submitted</li> </ul> |
|                                       | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No implementation of the QR code <i>generation</i> module</li> </ul>                                                                                                                                                                                                                                                                                                                 |



|                                             |                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <ul style="list-style-type: none"> <li>No implementation of reporting or statistics generation</li> <li>No front-end development</li> </ul>                                                                                                                                                                                                                               |
|                                             | <b>WP Result</b> <ul style="list-style-type: none"> <li>Robust, unit-tested <b>Backend API endpoints</b> for vote validation and submission</li> <li>A verified, tamper-proof core voting logic module</li> </ul>                                                                                                                                                         |
|                                             | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Token validation and eligibility logic complete</li> <li>Vote submission and immutability logic complete and unit-tested</li> </ul>                                                                                                                                                                    |
| <b>4.1<br/>Voting Interface Development</b> | <b>WP Content</b> <ul style="list-style-type: none"> <li>Develop the front-end UI for the voting process based on approved mockups (WP 2.1)</li> <li>Implement the QR code scanning functionality, including handling of webcam access and error states</li> <li>Integrate the UI with the core backend API endpoints (WP 3.2) to send and receive voting data</li> </ul> |
|                                             | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No development of the Admin Panel UI (WP 4.2)</li> <li>No implementation of security features like Audit Logging</li> </ul>                                                                                                                                                                                  |
|                                             | <b>WP Result</b> <ul style="list-style-type: none"> <li>A fully functional, responsive <b>Student Voting Application</b> front-end</li> <li>Successful end-to-end test of the "QR scan to vote submission" flow</li> </ul>                                                                                                                                                |
|                                             | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Completion of all static UI components</li> <li>Full API integration and successful functional testing</li> </ul>                                                                                                                                                                                      |
| <b>4.2<br/>Admin Interface Development</b>  | <b>WP Content</b> <ul style="list-style-type: none"> <li>Develop the front-end UI for the Admin Panel based on approved mockups (WP 2.2)</li> <li>Implement the core interfaces for candidate management and basic election controls</li> <li>Integrate UI elements with future backend endpoints (WP 5.1, 5.2)</li> </ul>                                                |
|                                             | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No development of the student voting interface (WP 4.1)</li> <li>No complex visualization implementation for reporting yet</li> </ul>                                                                                                                                                                        |
|                                             | <b>WP Result</b> <ul style="list-style-type: none"> <li>A functional, navigable <b>Admin Panel Front-End Application</b></li> <li>Basic data display and administrative interaction interfaces integrated</li> </ul>                                                                                                                                                      |
|                                             | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Completion of static UI components and dashboard layout</li> <li>Integration of all basic Admin Panel features</li> </ul>                                                                                                                                                                              |
| <b>5.1<br/>QR Generation Implementation</b> | <b>WP Content</b> <ul style="list-style-type: none"> <li>Develop the backend logic to generate unique, <b>encrypted</b> voting tokens/QR codes</li> <li>Implement the Admin Panel feature to <b>generate tokens in bulk</b></li> <li>Develop the secure <b>PDF export functionality</b> for printing the QR codes on individual or class sheets</li> </ul>                |
|                                             | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No implementation of QR code <i>scanning</i> (WP 4.1)</li> <li>No security audits on the encryption strength (WP 6.2)</li> </ul>                                                                                                                                                                             |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <b>WP Result</b> <ul style="list-style-type: none"> <li>The secured, functional <b>QR Code Generation Module</b></li> <li>The Admin Panel feature for bulk QR generation and tested PDF export</li> </ul>                                                                                                                                                                               |
|                                                  | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Token encryption and generation logic completed</li> <li>PDF export functionality completed and integrated into the Admin Panel</li> </ul>                                                                                                                                                                           |
| <b>5.2<br/>Evaluation &amp; Export Functions</b> | <b>WP Content</b> <ul style="list-style-type: none"> <li>Develop the <b>data aggregation logic</b> for detailed statistics</li> <li>Implement data visualization within the Admin Panel UI</li> <li>Develop and test the export functions for <b>CSV/Excel</b> and <b>PDF Report</b> formats</li> </ul>                                                                                 |
|                                                  | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No general security feature development (WP 5.3)</li> <li>No user training on how to use the reports (WP 7.2)</li> </ul>                                                                                                                                                                                                   |
|                                                  | <b>WP Result</b> <ul style="list-style-type: none"> <li>A fully functional <b>Reporting Dashboard</b> in the Admin Panel</li> <li>Validated and accurate generation of CSV/Excel and formatted PDF reports</li> </ul>                                                                                                                                                                   |
|                                                  | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Data aggregation logic and chart development complete</li> <li>Export functionality for both CSV/Excel and PDF formats fully tested</li> </ul>                                                                                                                                                                       |
| <b>5.3<br/>Security Implementation</b>           | <b>WP Content</b> <ul style="list-style-type: none"> <li>Implement the <b>election time window</b> control mechanism (start/stop API) for admins</li> <li>Develop and integrate the <b>Audit Log</b> system to record all critical actions</li> <li>Implement and test the basic <b>Manipulation Detection</b> logic</li> <li>Configure the database <b>Backup Mechanism</b></li> </ul> |
|                                                  | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No official third-party security audit (WP 6.2)</li> <li>No UI development (WP 4.2)</li> </ul>                                                                                                                                                                                                                             |
|                                                  | <b>WP Result</b> <ul style="list-style-type: none"> <li>An operational and robust <b>Audit Log system</b></li> <li>Tested election time control and automated backup functionality</li> </ul>                                                                                                                                                                                           |
|                                                  | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>Audit Log and Time Window implemented</li> <li>Backup and Manipulation Detection logic implemented and internally verified</li> </ul>                                                                                                                                                                                |
| <b>6.1<br/>Internal Tests</b>                    | <b>WP Content</b> <ul style="list-style-type: none"> <li>Perform <b>functional testing</b> of all features against the requirements document</li> <li>Conduct formal <b>usability tests</b> with a closed beta group to identify user friction points</li> <li>Document and prioritize all functional bugs and usability issues found in a central bug tracking system</li> </ul>       |
|                                                  | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>No security-specific penetration testing (WP 6.2)</li> <li>No code development or bug fixing (purely testing and reporting)</li> </ul>                                                                                                                                                                                     |
|                                                  | <b>WP Result</b> <ul style="list-style-type: none"> <li>A detailed <b>Bug and Issue Report Log</b></li> <li>A <b>Usability Test Report</b> with clear recommendations for improvement</li> </ul>                                                                                                                                                                                        |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>Completion of all functional test cases</li> <li>Completion of usability tests and final bug documentation sign-off</li> </ul>                                                                                                                                                                                         |
| <b>6.2</b><br><b>Security Audits</b>               | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>Engage a security expert (internal or external) to perform a <b>penetration test</b> on the system</li> <li>Verify the integrity and immutability of the stored vote records</li> <li>Fix and re-test all critical security vulnerabilities identified in the audit</li> </ul>                                                   |
|                                                    | <p><b>Non-WP Content</b></p> <ul style="list-style-type: none"> <li>No general functional testing (WP 6.1)</li> <li>No documentation work (WP 6.3)</li> </ul>                                                                                                                                                                                                                                    |
|                                                    | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>The <b>Final Security Audit Report</b></li> <li>System verified as secure</li> </ul>                                                                                                                                                                                                                                              |
|                                                    | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>Initial security audit and report generation</li> <li>Fixing of all critical issues and final security approval</li> </ul>                                                                                                                                                                                             |
| <b>6.3</b><br><b>Documentation</b>                 | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>Create a detailed <b>Admin User Guide</b></li> <li>Create a simple, accessible <b>Student Voter Guide</b></li> <li>Create the <b>Technical Handover Document</b></li> </ul>                                                                                                                                                      |
|                                                    | <p><b>Non-WP Content</b></p> <ul style="list-style-type: none"> <li>No further code development or bug fixes</li> <li>No training or deployment activities</li> </ul>                                                                                                                                                                                                                            |
|                                                    | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>Finalized and approved <b>Admin User Guide</b> and <b>Voter Guide</b></li> <li>A complete <b>Technical Documentation Package</b></li> </ul>                                                                                                                                                                                       |
|                                                    | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>Drafting of all three documentation pieces</li> <li>Final internal review, editing, and official approval of all documentation</li> </ul>                                                                                                                                                                              |
| <b>7.1</b><br><b>Prepare</b><br><b>Publishment</b> | <p><b>WP Content</b></p> <ul style="list-style-type: none"> <li>Prepare and finalize the production server/hosting environment</li> <li>Deploy the final, tested <b>VoteVox</b> application to the production environment</li> <li>Perform final <b>tests</b> in the live environment to ensure functionality</li> <li>Set up and verify ongoing monitoring and error logging systems</li> </ul> |
|                                                    | <p><b>Non-WP Content</b></p> <ul style="list-style-type: none"> <li>No final project sign-off (WP 7.2)</li> <li>No user training</li> </ul>                                                                                                                                                                                                                                                      |
|                                                    | <p><b>WP Result</b></p> <ul style="list-style-type: none"> <li>A <b>Live, fully operational VoteVox system</b> accessible to users</li> <li>Monitoring and logging systems successfully configured and running</li> </ul>                                                                                                                                                                        |
|                                                    | <p><b>Progress Measurement</b></p> <ul style="list-style-type: none"> <li>Deployment to production environment</li> <li>Successful test verification and monitoring setup completion</li> </ul>                                                                                                                                                                                                  |

|                                            |                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            |                                                                                                                                                                                                                                                                                |
| <b>7.2<br/>Training &amp;<br/>Handover</b> | <b>WP Content</b> <ul style="list-style-type: none"> <li>• Develop and deliver a comprehensive training session for the Election Commission/Administrators on how to operate and maintain the system</li> <li>• Formally hand over all final documentation (WP 6.3)</li> </ul> |
|                                            | <b>Non-WP Content</b> <ul style="list-style-type: none"> <li>• No further code development or bug fixes</li> </ul>                                                                                                                                                             |
|                                            | <b>WP Result</b> <ul style="list-style-type: none"> <li>• A <b>Trained and competent administrative staff</b></li> <li>• Formal handover and sign-off on knowledge transfer and documentation</li> </ul>                                                                       |
|                                            | <b>Progress Measurement</b> <ul style="list-style-type: none"> <li>• Completion of all training sessions</li> <li>• Formal client sign-off of knowledge transfer and support agreement</li> </ul>                                                                              |

## 1.10 Project Responsibility Matrix

| <div> <div>VoteVox<br/>1.4</div> <div>PROJECT-RESPONSIBILITY-MATRIX</div> </div> |                                           |                     |                 |                |                |
|----------------------------------------------------------------------------------|-------------------------------------------|---------------------|-----------------|----------------|----------------|
| WBS-Code                                                                         | WP-Title                                  | Roles & Environment |                 |                |                |
|                                                                                  |                                           | Project Owner       | Project Manager | Project Team A | Project Team B |
| <b>0</b>                                                                         | <b>Project Management</b>                 |                     |                 |                |                |
| 0.1                                                                              | Coordinate Project                        | A                   | R               | O              | O              |
| 0.2                                                                              | Control Project                           | A                   | R               | O              | O              |
| <b>1</b>                                                                         | <b>Project Planning</b>                   |                     |                 |                |                |
| 1.1                                                                              | Requirements Analysis                     | A                   | R               | R              | R              |
| <b>2</b>                                                                         | <b>Design &amp; UI Development</b>        |                     |                 |                |                |
| 2.1                                                                              | Voting Interface Design                   | A                   | R               | S              | S              |
| 2.2                                                                              | Admin Interface Design                    | A                   | S               | R              | R              |
| <b>3</b>                                                                         | <b>Backend Development &amp; Database</b> |                     |                 |                |                |
| 3.1                                                                              | Database Setup                            | A                   | A               | R              | S              |
| 3.2                                                                              | Logic Implementation                      | A                   | R               | S              | R              |
| <b>4</b>                                                                         | <b>Frontend Development</b>               |                     |                 |                |                |
| 4.1                                                                              | Voting Interface Development              | A                   | R               | S              | S              |
| 4.2                                                                              | Admin Interface Development               | A                   | S               | R              | R              |
| <b>5</b>                                                                         | <b>Features &amp; Security</b>            |                     |                 |                |                |
| 5.1                                                                              | QR Generation Implementation              | A                   | S               | R              | S              |
| 5.2                                                                              | Evaluation & Export Functions             | A                   | S               | S              | R              |
| 5.3                                                                              | Security Implementation                   | A                   | R               | S              | S              |
| <b>6</b>                                                                         | <b>Testing &amp; Documentation</b>        |                     |                 |                |                |
| 6.1                                                                              | Internal Tests                            | A                   | S               | R              | S              |
| 6.2                                                                              | Security Audits                           | A                   | S               | S              | R              |
| 6.3                                                                              | Documentation                             | A                   | R               | S              | S              |
| <b>7</b>                                                                         | <b>Deployment &amp; Handover</b>          |                     |                 |                |                |
| 7.1                                                                              | Prepare Publishment                       | A                   | R               | R              | R              |
| 7.2                                                                              | Training & Handover                       | A                   | R               | R              | R              |

### Responsibility Levels

- o **R (Responsible):** The person who does the work to complete the task
- o **A (Accountable):** The person ultimately accountable for the correct completion of the task
- o **S (Support):** Provides resources or plays a supporting role in implementation
- o **O (Out of the Loop):** Not involved in the task

## 1.11 Milestoneplan

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### MILESTONEPLAN

| WBS-Code | Milestone                    | Plan date  | Revised date | Actual date |
|----------|------------------------------|------------|--------------|-------------|
| 0        | Project Start                | 12.03.2026 |              |             |
| 1        | Concept Approval             | 19.03.2026 |              |             |
| 2        | UI Design Sign-off           | 26.03.2026 |              |             |
| 3        | Core Function Implementation | 02.04.2026 |              |             |
| 4        | Frontend Integration         | 09.04.2026 |              |             |
| 5        | Feature Complete             | 30.04.2026 |              |             |
| 6        | System Acceptance            | 07.05.2026 |              |             |
| 7        | Project Completion           | 13.05.2026 |              |             |

\*In order of plan dates.

## 1.12 Project Bar Chart



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pp

## 1.13 Resource Plan

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### Resource Plan

| WBS-Code | Phase/Work-package            | Type of resource                                 | Planned quantity (days) | Revised quantity (days) | Actual quantity (days) | Deviation (days) |
|----------|-------------------------------|--------------------------------------------------|-------------------------|-------------------------|------------------------|------------------|
| 0.1      | Coordinate Project            | Hubauer Paul<br>Matthias                         | 45                      |                         |                        |                  |
| 0.2      | Control Project               | Hubauer Paul<br>Matthias                         | 45                      |                         |                        |                  |
| 1.1      | Requirements Analysis         | Team                                             | 3                       |                         |                        |                  |
| 2.1      | Voting Interface Design       | Hubauer Paul<br>Matthias                         | 3                       |                         |                        |                  |
| 2.2      | Admin Interface Design        | Anzengruber Laurenz,<br>Aigner Sandro            | 3                       |                         |                        |                  |
| 3.1      | Database Setup                | Aigner Sandro                                    | 3                       |                         |                        |                  |
| 3.2      | Logic Implementation          | Hubauer Paul<br>Matthias,<br>Anzengruber Laurenz | 3                       |                         |                        |                  |
| 4.1      | Voting Interface Development  | Hubauer Paul<br>Matthias                         | 3                       |                         |                        |                  |
| 4.2      | Admin Interface Development   | Anzengruber Laurenz,<br>Aigner Sandro            | 3                       |                         |                        |                  |
| 5.1      | QR Generation Implementation  | Aigner Sandro                                    | 13                      |                         |                        |                  |
| 5.2      | Evaluation & Export Functions | Anzengruber Laurenz                              | 13                      |                         |                        |                  |
| 5.3      | Security Implementation       | Hubauer Paul<br>Matthias                         | 13                      |                         |                        |                  |
| 6.1      | Internal Tests                | Aigner Sandro                                    | 3                       |                         |                        |                  |
| 6.2      | Security Audits               | Anzengruber Laurenz                              | 3                       |                         |                        |                  |
| 6.3      | Documentation                 | Hubauer Paul<br>Matthias                         | 3                       |                         |                        |                  |
| 7.1      | Prepare Publishment           | Team                                             | 1                       |                         |                        |                  |
| 7.2      | Training & Handover           | Team                                             | 4                       |                         |                        |                  |



## 1.14 Project Cost Plan

| <div> VoteVox<br/>1.4 </div> PROJECT COST PLAN |                     |                 |                 |             |           |
|------------------------------------------------|---------------------|-----------------|-----------------|-------------|-----------|
| WBS-Code,<br>WP-Title                          | Type of Cost        | Planned<br>cost | Revised<br>cost | Actual cost | Deviation |
|                                                | • Personnel         |                 |                 |             |           |
|                                                | • Material          |                 |                 |             |           |
|                                                | • External services |                 |                 |             |           |
|                                                | • Other             |                 |                 |             |           |
|                                                | Total               |                 |                 |             |           |
|                                                | • Personnel         |                 |                 |             |           |
|                                                | • Material          |                 |                 |             |           |
|                                                | • External services |                 |                 |             |           |
|                                                | • Other             |                 |                 |             |           |
|                                                | Total               |                 |                 |             |           |
|                                                | • Personnel         |                 |                 |             |           |
|                                                | • Material          |                 |                 |             |           |
|                                                | • External services |                 |                 |             |           |
|                                                | • Other             |                 |                 |             |           |
|                                                | Total               |                 |                 |             |           |
| Project cost                                   |                     |                 |                 |             |           |

## 1.15 Project Communication

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1.4

### PROJECT- COMMUNICATION

| Title                  | Objectives, Content                                                                                                                                                                                             | Participants                                 | Schedule                  | Location  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------|-----------|
| Team meeting           | <ul style="list-style-type: none"> <li>• Project status</li> <li>• Important technical decisions</li> <li>• Review of changes in requirements</li> <li>• Update development and testing timeline</li> </ul>     | Project manager, Project team members        | First workday of the week | Classroom |
| Morning meeting        | <ul style="list-style-type: none"> <li>• Current project status</li> <li>• Discuss daily work packages</li> <li>• Plan testing activities or special events (election day, distribution schedule)</li> </ul>    | Project manager, Project team, Project coach | 7:30 AM every workday     | Classroom |
| Administration meeting | <ul style="list-style-type: none"> <li>• Communicate project status to customer</li> <li>• Discuss requirement changes</li> <li>• Clarify data protection</li> <li>• Discuss delays and their causes</li> </ul> | Customer, Project manager                    | Last workday of the month | Classroom |

## 1.16 Project „Rules“

- **1.16.1 General**
  - Team members must maintain a respectful and professional working environment.
    - Slurs or insults are fined with a donation to the coffee cash register.
  - The official project language is English.
    - All communication and documentation must be written in English.
- **1.16.2 Communication**
- **Communication Platform**
  - **Formal communication:**
    - Any important changes, issues, or questions regarding the project must be communicated via Email to the project manager or customer.
  - **Informal communication:**
    - For quick coordination and team-internal questions, we use Microsoft Teams, WhatsApp, and similar platforms.
- **Meetings**
  - **Weekly team meetings**
    - Discuss project status, important decisions, timeline updates.
  - **Morning meetings**
    - Clarify daily tasks, coordinate testing, short status update.
  - **Monthly customer meetings**
    - Present current progress, discuss requirement changes and delays.
- **1.16.3 Task Management**
  - Work packages must be followed strictly.
  - Any delays must be reported immediately.
- **1.16.4 Development**
  - The code must be maintained clean and have a readable project structure.
  - The system must be optimized to meet the project goals:
    - Secure QR code generation
    - Reliable vote validation
    - Clear separation between identity and vote data
    - User-friendly interface for students and staff
  - All features must be tested before merging into the main branch.
- **1.16.5 Version Control & Documentation**
  - Git is used for version control.
    - All commits must contain clear and meaningful messages.
  - Documentation must always be kept up to date, including technical documents, meeting results, and system changes.
- **1.16.6 Data Protection**
  - All user inputs must be validated to prevent bugs and manipulation attempts.
  - Only minimal necessary data is stored - no personal data beyond eligibility information.
  - Identity and vote data must always remain strictly separated (GDPR compliant).
  - Only authorized team members may access sensitive information.
  - Backup files must be encrypted and stored securely.

## 1.17 Project Risk Analysis

| PROJECT RISK ANALYSIS |                               |                                                 |           |               |              |            |          |                                      |                          |
|-----------------------|-------------------------------|-------------------------------------------------|-----------|---------------|--------------|------------|----------|--------------------------------------|--------------------------|
| WBS-Code              | WBS-description               | Risk-description                                | Priority  | Benefit /Cost | Probability  | Risk-value | Delay    | Avoiding action                      | Cost for minimizing risk |
| (Code)                | (Text)                        | (Text)                                          | (Range)   | (Euro)        | (Percentage) | (Euro)     | (iWeeks) | (Text)                               | (Euro)                   |
| 0.1                   | Coordinate Project            | Miscommunication between departments            | Medium    |               | 20%          |            | 1        | Weekly sync meetings                 |                          |
| 0.2                   | Control Project               | Scope changes during project                    | High      |               | 30%          |            | 2        | Strict change management             |                          |
| 1.1                   | Requirements Analysis         | Incomplete requirements                         | High      |               | 25%          |            | 3        | Stakeholder workshops                |                          |
| 2.1                   | Voting Interface Design       | Poor usability = invalid votes                  | High      |               | 30%          |            | 2        | UI prototypes & testing              |                          |
| 2.2                   | Admin Interface Design        | Security vulnerabilities in concept             | High      |               | 20%          |            | 2        | Security review                      |                          |
| 3.1                   | Database Setup                | Incorrect election logic                        | Very High |               | 15%          |            | 4        | Logic validation by experts          |                          |
| 3.2                   | Logic Implementation          | Data loss or corruption                         | High      |               | 20%          |            | 3        | Backups & recovery plan              |                          |
| 4.1                   | Voting Interface Development  | Device compatibility issues                     | Medium    |               | 25%          |            | 2        | Cross-device testing                 |                          |
| 4.2                   | Admin Interface Development   | Unauthorized access                             | Very High |               | 30%          |            | 3        | Multi-factor authentication          |                          |
| 5.1                   | QR Generation Implementation  | Unreadable QR codes                             | Medium    |               | 20%          |            | 2        | Test QR on various printers/devices  |                          |
| 5.2                   | Evaluation & Export Functions | Incorrect data export                           | Medium    |               | 20%          |            | 2        | Automated export validation          |                          |
| 5.3                   | Security Implementation       | Cyber-attack during election                    | Very High |               | 10%          |            | 4        | Penetration tests & monitoring       |                          |
| 6.1                   | Internal Tests                | Low test coverage                               | High      |               | 25%          |            | 3        | Increase coverage + regression tests |                          |
| 6.2                   | Security Audits               | Compliance issues                               | Medium    |               | 10%          |            | 2        | External security checks             |                          |
| 6.3                   | Documentation                 | Insufficient documentation = Maintenance issues | Medium    |               | 20%          |            | 1        | Documentation review                 |                          |
| 7.1                   | Prepare Publishment           | Deployment failure                              | High      |               | 15%          |            | 2        | Backup deployment strategy           |                          |
| 7.2                   | Training & Handover           | User misoperation                               | Medium    |               | 25%          |            | 2        | Training material & sessions         |                          |
| Project total         |                               |                                                 |           |               |              |            |          |                                      |                          |

## 1.18 Project Documentation

| Area                 | Description                                                                                                                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File                 |  <br>VoteVox_Work_Break down_Structure.pdf    VoteVox_BarChart.mpp                         |
| Access Authorisation | Hubauer Paul Matthias, Anzengruber Laurenz, Aigner Sandro                                                                                                                                                                                                    |
| Naming convention    | VoteVox_Taskname.extension                                                                                                                                                                                                                                   |
| Rules                | <p>Every Monday a report stating what was done and changed in the last week and which goals were achieved or not achieved.</p> <p>Every Monday a report stating what was done and changed in the last week and which goals were achieved or not achieved</p> |

## 2 Project Start

### **2.1 Minutes– Project Start**

#### **2.1.1 Project Start Workshop**

#### **2.1.2 Follow-up Workshop**

#### **2.1.3 Project Owner Meeting**

## 3 Project Co-ordination

### 3.1 Approval of Work-packages

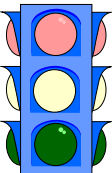
| <div> VoteVox<br/>1.4 </div> <div> APPROVAL OF<br/>WORK-PACKAGES </div> |                               |                          |                                |             |           |
|-------------------------------------------------------------------------|-------------------------------|--------------------------|--------------------------------|-------------|-----------|
| WBS-Code                                                                | Work-package                  | WP-Owner                 | Date                           | Approval by | Signature |
| 0.1                                                                     | Coordinate Project            | Hubauer Paul<br>Matthias | 13 <sup>th</sup> May<br>2026   |             |           |
| 0.2                                                                     | Control Project               | Hubauer Paul<br>Matthias | 13 <sup>th</sup> May<br>2026   |             |           |
| 1.1                                                                     | Requirement Analysis          | Hubauer Paul<br>Matthias | 15 <sup>th</sup> March<br>2026 |             |           |
| 2.1                                                                     | Voting Interface Design       | Hubauer Paul<br>Matthias | 22 <sup>nd</sup> March<br>2026 |             |           |
| 2.2                                                                     | Admin Interface Design        | Aigner Sandro            | 22 <sup>nd</sup> March<br>2026 |             |           |
| 3.1                                                                     | Database Setup                | Aigner Sandro            | 29 <sup>th</sup> March<br>2026 |             |           |
| 3.2                                                                     | Logic Implementation          | Hubauer Paul<br>Matthias | 29 <sup>th</sup> March<br>2026 |             |           |
| 4.1                                                                     | Voting Interface Development  | Hubauer Paul<br>Matthias | 5 <sup>th</sup> April<br>2026  |             |           |
| 4.2                                                                     | Admin Interface Development   | Aigner Sandro            | 5 <sup>th</sup> April<br>2026  |             |           |
| 5.1                                                                     | QR Generation Implementation  | Aigner Sandro            | 26 <sup>th</sup> April<br>2026 |             |           |
| 5.2                                                                     | Evaluation & Export Functions | Anzengruber<br>Laurenz   | 26 <sup>th</sup> April<br>2026 |             |           |
| 5.3                                                                     | Security Implementation       | Hubauer Paul<br>Matthias | 26 <sup>th</sup> April<br>2026 |             |           |
| 6.1                                                                     | Internal Tests                | Aigner Sandro            | 3 <sup>rd</sup> May<br>2026    |             |           |
| 6.2                                                                     | Security Audits               | Anzengruber<br>Laurenz   | 3 <sup>rd</sup> May<br>2026    |             |           |
| 6.3                                                                     | Documentation                 | Hubauer Paul<br>Matthias | 3 <sup>rd</sup> May<br>2026    |             |           |
| 7.1                                                                     | Prepare Publishment           | Hubauer Paul<br>Matthias | 7 <sup>th</sup> May<br>2026    |             |           |
| 7.2                                                                     | Training & Handover           | Hubauer Paul<br>Matthias | 13 <sup>th</sup> May<br>2026   |             |           |

## 3.2 Minutes – Project Co-ordination



## 4 Project Controlling

### 4.1 Project Status Report

| <div style="display: flex; justify-content: space-between;"> <div>           &lt;Project name&gt;<br/>           &lt;Project no.&gt; </div> <div> <b>PROJECT STATUS REPORT</b><br/>           as of ..... </div> </div> |                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  <div style="margin-left: 10px;"> <p>Project crisis</p> <p>Project in difficulties</p> <p>Project according to plan ✓</p> </div>       | <b>1) Overall Status:</b><br><ul style="list-style-type: none"> <li></li> </ul>    |
| <b>2) Status Project objectives</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                     | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |
| <b>3) Status Project progress</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                       | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |
| <b>4) Status Schedule</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                               | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |
| <b>5) Status Resources/ costs</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                       | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |
| <b>6) Status Context</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                                | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |
| <b>7) Status Organisation/ culture</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                  | <b>Activities:</b><br><ul style="list-style-type: none"> <li></li> <li></li> </ul> |

## **4.2 Additional Project Status Reports**

## **4.3 Minutes – Project controlling**

### **4.3.1 Project Controlling Meetings**

### **4.3.2 Project Owner Meetings**

# 5 Project Close Down

## 5.1 Project Close Down Report

| <div style="display: flex; justify-content: space-between;"> <div>           &lt;Project name&gt;<br/>           &lt;Project no.&gt; </div> <div style="text-align: center;"> <b>PROJECT CLOSE DOWN<br/>REPORT</b> </div> </div>                                                                                                                                                                                                                                                                     |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------|-------|-------|----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 1) Overall impression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       | 2) Reflection: Fulfilment of objectives                                     |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 3) Reflection: Deliverables / Schedule                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 4) Reflection: Resources / Costs                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 5) Reflection: Internal Organisation / Environmental Relationships                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 6) Performance appraisal (Project owner, Project manager, Project member)                                                                                                                                                                                                                                                                                                                                                                                                                            |       | 7) Lessons learned (Summary of Experiences and suggestions for improvement) |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>8) Post-Project Phase Planning, Additional Tasks</b> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 60%;">To-Do</th> <th style="width: 20%;">Owner</th> <th style="width: 20%;">Schedule</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> </tbody> </table> |       |                                                                             | To-Do | Owner | Schedule |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| To-Do                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Owner | Schedule                                                                    |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>9) Project Close Down</b> <div style="display: flex; justify-content: space-around; margin-top: 20px;"> <div style="text-align: center;">           .....<br/>           &lt;Name&gt; (Project owner)         </div> <div style="text-align: center;">           .....<br/>           &lt;Name&gt; (Project manager)         </div> </div>                                                                                                                                                        |       |                                                                             |       |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

## **5.2 Minutes – Project Close Down**

### **5.2.1 Project Close Down Workshop**